

Project Problem and Success Test

Project: Transparent Book Recommender

1. Your mission

Your learning target

I can define a small application purpose.

I can turn requirements into checkable acceptance cases.

I can plan a feasible pipeline and human review.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Permissions and Human Review. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Evidence and Sources. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Image and Voice Pipelines. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 32: Project Problem and Success Test

A project is defined by user need and measurable success, not feature count.



THINKER MISSION

Write a recommender brief and three acceptance tests.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Why must a recommendation show no eligible option when every item fails a hard constraint?

Readiness for A reproducible data investigation

What should happen to an invalid duration instead of silently making it zero?

My readiness explanation

This week's end goal

By the end, I can plan a recommender with acceptance tests before writing its selection code.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use project problem and success test to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

Plan a fictional activity chooser for a teacher-supervised 20-minute session. It will validate a small local dataset, exclude over-limit items, rank eligible items using $2 * \text{fun} + \text{learning}$ and break ties alphabetically. It will explain the result for human review. It will not book, message or collect personal student profiles.

Apply the connected idea: A reproducible data investigation

Investigate the fictional question: among supplied sessions with valid duration records, what is the mean duration? Inputs are 10, 0, 20, None and -2 minutes. Our schema accepts exact integers 0 through 60. The wording deliberately limits the answer to accepted records.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



IN

INPUT

Write a recommender brief and three acceptance tests.



PROCESS

A project is defined by user need and measurable success, not feature count.



OUT

OUTPUT

Write a recommender brief and three acceptance tests.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

```
requirements = {"limit": 20, "fun_weight": 2,
                "learning_weight": 1, "tie": "name ascending"}
print(requirements["limit"])
print(requirements["tie"])
```

Follow the evidence

Step or case · Evidence · Meaning

Step or case: Input

Evidence: Local fictional records

Meaning: Validate schema

Step or case: Decision

Evidence: Filter then score then tie rule

Meaning: Explain selected item

Step or case: Output

Evidence: Suggestion or no eligible item

Meaning: Teacher reviews

Read each row from left to right and connect the evidence to the stated rule.

Define fields name, minutes, fun and learning. Require a nonempty text name, integer minutes 0 through 60 and integer ratings 0 through 5. Reject duplicate names because the tie rule uses names as identifiers. Write cases before implementing so requirements do not drift to fit outputs.

Turn a wish into a checkable plan

A phrase such as make a smart choice does not say what counts as success. The chooser plan does: validate the named fields, include durations at most 20 minutes, calculate 2 times fun plus learning, and use alphabetical names to break a tie. Describe the expected output and a no-match result before coding. Another learner should be able to use your plan to predict the same answer without guessing what you intended.

Apply the connected idea: A reproducible data investigation

```
raw = [10, 0, 20, None, -2]

valid = []

rejected = 0

for value in raw:

    if type(value) is int and 0 <= value <= 60:

        valid.append(value)

    else:

        rejected = rejected + 1

print(len(valid), rejected)

if len(valid) > 0:

    print(sum(valid) / len(valid))
```

```
else:  
    print("No valid observations")
```

Follow the evidence

Step or case · Evidence · Meaning

Step or case: Raw evidence

Evidence: Five supplied values

Meaning: Keep unchanged

Step or case: Validation

Evidence: Accept 10, 0, 20

Meaning: Reject unknown and negative

Step or case: Summary

Evidence: 30 divided by 3

Meaning: Mean 10 minutes

Read each row from left to right and connect the evidence to the stated rule.

The result is 3 accepted, 2 rejected, and mean 10.0 minutes. A reproducible report states the question, raw input version, schema, exclusion reasons, calculation and scope. Someone else should obtain the same result from the same inputs and rules.

Keep the denominator with the evidence

First filter the raw durations using the written schema. Keep 10, 0 and 20; reject None because the duration is unknown and -2 because it is outside the range. The accepted sum is 30 and the accepted count is three, so the accepted-record mean is 10 minutes. Dividing by all five supplied records would treat the two rejected durations as if they contributed known zeros. State three accepted out of five beside the mean.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL
Expected everyday case



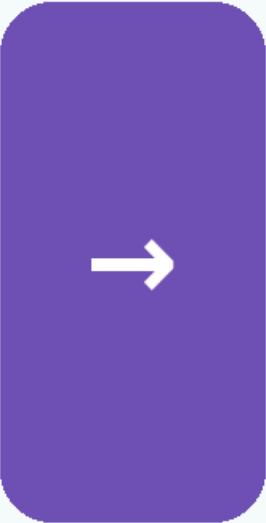
BOUNDARY
Near a decision edge

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING

Needed information absent



TRANSFER

Fresh situation or user

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: Reduce the limit from 8 to 5.

Prediction: Return no eligible item.

Reason: All three valid rows take more than five minutes.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

REQUIREMENT

behavior the solution must provide

3

ACCEPTANCE TEST

observable pass/fail check

4

SCOPE

what is and is not included

Word · My own example · Not this / boundary

Word: user need

My own example: The problem or desired outcome the system is meant to address for a person.

Not this / boundary: _____

Word: requirement

My own example: A behaviour or condition the proposed system must satisfy.

Not this / boundary: _____

Word: acceptance test

My own example: A predefined check that a product satisfies a stated requirement.

Not this / boundary: _____

Word: scope

My own example: The tasks, users, information and actions included within an agreement or design.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Acceptance case

Meaning: An input and expected behavior used to judge a requirement

Word: Non-goal

Meaning: Something intentionally outside project scope

Word: Prototype

Meaning: A limited implementation for testing a design

Terms for A reproducible data investigation

Term · Meaning

Term: Investigation

Meaning: A documented process for answering a question with evidence

Term: Pipeline

Meaning: An ordered sequence of processing steps

Term: Audit trail

Meaning: A record of inputs, decisions and changes

Term: Limitation

Meaning: A reason a conclusion has restricted scope

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 32 • WORKED EVIDENCE

- 1 Need: help a reader choose from 12 classroom books.
- 2 Requirements: use stated preferences; show score; protect privacy.
- 3 Tests: correct arithmetic; unsuitable hard constraint removed; explanation names evidence.
- 4 Out of scope: claiming the objectively 'best' book or storing private history.

- 1. Need: help a reader choose from 12 classroom books.
- 2. Requirements: use stated preferences; show score; protect privacy.
- 3. Tests: correct arithmetic; unsuitable hard constraint removed; explanation names evidence.
- 4. Out of scope: claiming the objectively 'best' book or storing private history.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Write an acceptance case for the duration boundary and one for no eligible option.

Apply the connected idea: A reproducible data investigation

If another valid value 30 is added, calculate the new accepted count and mean.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Project Problem and Success Test

A fictional reading-break chooser needs a transparent contract.

The facts we will use

Items have id text, minutes integer 0–30, interest integer 0–5. Request limit is integer 0–30.

Eligible if minutes≤limit; score=interest; highest wins; ties choose alphabetically smaller ID; invalid rows excluded and counted.

Project Problem and Success Test

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Specify inputs

Validate each row and request; keep rejected-row reasons.



2. Filter and rank

At limit 8: A(time8,interest3), B(9,5), C(6,3). B is ineligible.



3. Expected output

A beats tied C by ID; report A and its score 3.



4. Acceptance cases

Also plan invalid request, no eligible item and equal-score cases.

CHANGE ONE THING

Reduce the limit from 8 to 5.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Specify inputs

What happens and why: Validate each row and request; keep rejected-row reasons.

Step: 2. Filter and rank

What happens and why: At limit 8: A(time8,interest3), B(9,5), C(6,3). B is ineligible.

Step: 3. Expected output

What happens and why: A beats tied C by ID; report A and its score 3.

Step: 4. Acceptance cases

What happens and why: Also plan invalid request, no eligible item and equal-score cases.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Planning fixes what success means before code returns an answer. The duration boundary is including equality, and the tie policy prevents accidental list-order dependence. A complete output may include why other rows were excluded. Test expectations should come from the contract rather than copying the first program result.

What this example does not establish

The supplied interest scores are fictional; the specification does not establish genuine learner preferences.

Read the labelled picture: Write an equality test and explain which requirement it checks.

Change one thing: Reduce the limit from 8 to 5. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: A project is better when it has more features. Strong move: locate evidence, change one cause, and retest.

Evidence record • Expected • Actual • Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

The scope is a rule-based prototype using fictional ratings. It does not demonstrate personalized learning

impact or a trained AI model. A clear non-goal protects students from building an untestable or overly broad project.

Name two capabilities excluded from this prototype and explain why.

Apply the connected idea: A reproducible data investigation

Missing or invalid durations might belong to unusually long sessions. The accepted mean may therefore differ from the full group's mean. Do not make a causal claim from this descriptive investigation.

A report gives only mean 9. What additional evidence lets a peer reproduce it?

8. Practise together

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET evidence_record PREDICT expected_result RUN one normal case and one boundary case
COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST
EXPLAIN result, limitation, and next safe step

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

A requirement such as make a smart choice cannot be tested precisely. Replace it with a stated score, limit, tie rule and output explanation. If real feedback later changes the policy, record a versioned change rather than rewriting previous evidence.

Apply the connected idea: A reproducible data investigation

Dividing the accepted total 30 by all five records gives 6 and answers no clear observed-duration question. Use the accepted count for this mean, disclose exclusions, and never describe it as the average for every supplied session.

Practise with less help: Write an acceptance case

Use workbook W2 for the connected example “Planning the application”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Planning the application. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Practise the connected topic: A reproducible data investigation

Use its named workbook W2. Identify the given inputs, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A successful run proves the conclusion

Explain the picture

Write an equality test and explain which requirement it checks.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → TEST → RECORD → EXPLAIN

Interview a reader, write one need statement, define three measurable tests and two out-of-scope items, then peer-review for testability.

Interview a reader, write one need statement, define three measurable tests and two out-of-scope items, then peer-review for testability.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



MASTERY LADDER
1 Name it • 2 Trace it • 3 Apply it • 4 Challenge it • 5 Explain limits

1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or evidence path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



TRANSPARENT BOOK RECOMMENDER

Freeze the brief before building so later choices can be traced to requirements.

THIS WEEK'S CHECKPOINT Freeze the brief before building so later choices can be traced to requirements.

SYSTEM MAP Reader need → allowed inputs → transparent rule or score → ranked candidates → explanation → test record → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Retain the Transparent Book Recommender’s schema, constraints, score, tie rule and acceptance table as its build specification.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT PROOF

Write a recommender brief and three acceptance tests.

CHALLENGE Write a recommender brief and three acceptance tests.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 32 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A project recommends from ten approved books using genre and length.

The question

Define acceptance tests.

Worked explanation

Verify book existence, hard constraints, correct scoring explanations, tie handling and no-match behaviour on predefined requests.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Turn a project goal into acceptance tests

Evidence or stage · Value or result

Evidence or stage: Catalogue

Value or result: Suggestion must be one of ten approved books

Evidence or stage: Constraints

Value or result: Genre and length respected

Evidence or stage: Scoring

Value or result: Explanation matches the calculation

Evidence or stage: Boundaries

Value or result: Handle ties and no matches

A project recommends from ten approved books using genre and length.

Define acceptance tests.

Verify book existence, hard constraints, correct scoring explanations, tie handling and no-match behaviour on predefined requests.

14. Try a new case independently

Independent case: Plan independently

Plan a 15-minute fictional club chooser. Specify schema, score, duration limit, tie rule, invalid-data behavior, output and three acceptance cases.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 32 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Define a testable goal for a book recommender serving readers who want short space books. Include a concrete success condition.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Freeze the brief before building so later choices can be traced to requirements.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

Specify three acceptance tests for a recommender: a normal match, a tied score and no qualifying book. Include expected behaviour before implementation.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: A fictional reading-break chooser needs a transparent contract.

By the end, I can · How I will show it

By the end, I can: Define a small application purpose

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Turn requirements into checkable acceptance cases

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Plan a feasible pipeline and human review

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project

evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Define a small application purpose

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional reading-break chooser needs a transparent contract.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Also plan invalid request, no eligible item and equal-score cases.

1. Filter and rank
2. Specify inputs
3. Acceptance cases
4. Expected output

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Acceptance cases Also plan invalid request, no eligible item and equal-score cases. Planning fixes what success means before code returns an answer. The duration boundary is including equality, and the tie policy prevents accidental list-order dependence. A complete output may include why other rows were excluded. Test expectations should come from the contract rather than copying the first program result.

Practice 2: explain the deciding evidence

Changed-case practice: Reduce the limit from 8 to 5. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future input.
- 2. Return no eligible item.
- 3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Return no eligible item. All three valid rows take more than five minutes.

Repair a missing building block

Review Permissions and Human Review (ai-2/31). Goal: Explain the mechanism and terminology of least privilege and tool scopes. Short practice: Which proposal is allowed, and why is an authorised object alone insufficient?. Return to this week and explain what you changed.

Review Evidence and Sources (ai-2/29). Goal: Explain the mechanism and terminology of claim support and citation evaluation. Short practice: Why is the original answer not fully supported despite having citations on every claim?. Return to this week and explain what you changed.

Review Image and Voice Pipelines (ai-2/26). Goal: Explain the mechanism and terminology of speech

pipelines and multimodal evidence. Short practice: Which stage first fails, and why should the faulty transcript not be silently overwritten?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.